BT-403 MCQs Spring 2023

(Collected by work together team)

Rearrangement By M.Faryad

mid + final ;

1. Supplement the basal medium of **Murashige and Skoog (1962)**
2. **Breeding :**Breeding is more precise and efficient.• Artificially mating • Cross-pollination
3. In **1990** -The first food product of biotechnology (an enzyme used in Cheese production and a yeast used for baking) appeared on the Market.
4. In**1995**, farmers have been growing genetically engineered (**GE**) Crops.
5. In **2003**, **7** million farmers in **18** countries more than 85 percent of Them resource-poor farmers in the developing world were planting Biotech crops.
6. In a world whe**re 7 Billion** people, living mostly in rural areas, go Hungry every day,
7. Food demand is set to **double** in the next **thirty years a**nd arable land is Limited
8. Biotech crops may provide enhanced quality traits such as increased Levels of beta-carotene in rice to aid in reducing vitamin A deficiencies
9. •**Genetic engineering,** also called **genetic modification, is** the direct Manipulation of an organism's genes using biotechnology.
10. Inserts fragments of DNA into chromosomes of cells
11. Uses tissue culture to regenerate the cells into a whole organism with Different genetic composition from the original cells. • This is also known as **rDNA technology**
12. •One of the earliest protein based markers to be used was **Isozyme**.
13. Biotechnology-derived **vaccines** are used inLivestock and humans.
14. Tissue culture is the **regeneration** of plants in the laboratory from Disease-free plant parts.
15. This technique allows for the reproduction of **disease-free planting Material** for crops.
16. Examples of crops produced using tissue culture include **citrus, Pineapples, avocados, mangoes, bananas, coffee and papaya.**
17. Gene identification and transfer techniques used to improve the **color, Smell, size a**nd other features of flowers.
18. •**Golden Rice, which** contains **Beta-carotene**, the precursor for Vitamin A production In our bodies.

19. Pest resistant crops :• Pest resistant GM crops (**primarily cotton and Maize),**
20. Genetically modified so they are toxic to certain Insects. • They are often called **Bt crops**
21. The introduced genes were originally identified in a bacterial species called **Bacillus thuringiensis.**
22. •Tissue culture involves **both plant and animal cells .**
23. Tissue culture produces **clones**, in which all product cells have the **same Genotype** (unless affected by mutation during culture)
24. Tissue culture had its origins at the beginning of the **20th** century with The work of • Gottleib Haberlandt **(plants)** • Alexis Carrel (**animals**) •
25. The first commercial use of plant **clonal propagation** on **artificial Media** was in the germination and growth of **orchid plants**, in the **1920's•**
26. In the **1950's** and 60's there was a great deal of research, but it was Only after the development of a reliable artificial medium (**Murashige & Skoog, 1962)** that plant tissue culture really 'took off' commercially
27. A suitable growth medium containing **energy sources and inorganic Salts** to supply cell growth needs. • This can be **liquid or semisolid**
28. Growth regulators – in plants, both **auxins** & **cytokinins**.
29. **Explants** are small pieces of plant parts or tissues that are aseptically Cut and used to initiate a culture in a nutrient medium .
30. Explants can be taken from different parts of a plant such **as shoots, Leaves, stems, flowers, roots,** and from many types of mature cell
31. The shoots are transferred to a growth Medium with relatively **higher auxin: Cytokinin ratios •** Growth is a **continuous process** Development is phase to **phase process**
32. Plant Growth regulators (PGR) refers to natural or synthetic Substances influence the **growth and developmentNatural hormone:**
33. Produced by some tissues in the plant. Also called Endogenous Hormones. E.g. **IAA (Indol acetic acid) .• Synthetic hormone:**
34. Produced **artificially** and similar to natural hormone in physiological Activity. Also called **Exogenous** hormones.
35. Growth Inhibit the growth of plant. e.g. **ABA, Ethylene.**
36. Derived from the **Greek** word "**auxein**" means- "to **grow/increase"**.
37. Auxins are synthesized in the **stem and root apic**es and transported Through the plant axis. **Gibberellins** Stimulates stem growth dramatically
38. **Cytokinins** • Promotes cell division • Found in all tissues with considerable **cell division.**
39. May promote leaf senescing and abscission (**leaf fall).** Increases Female flowers in **cucumbers** (economically – will increase fruit Production. High-efficiency particulate air (**HEPA**) filter. .
40. A **16-**h light photoperiod is optimal for tissue cultures, and a Temperature of **22–258C** is used in most laboratories.
41. A light intensity of **25–50□mol.m-2s-**1 PAR (**photosynthetically active Radiation**) is typical for tissue cultures and is supplied by **cool white Fluorescent lamps.**
42. •A relative humidity of **50–60% is m**aintained in the growth chambers.

43. The important media used for all purpose Experiment are Murashige and Skoog medium **(MS medium)**
44. The culture medium is **closed with cotton plug**/ Or aluminium foil sheet.
45. The pH of the medium is adjusted **to 5.8 (acidic Range)**.
46. inorganic salts (both micro & macro elements) a carbon source **(usually sucrose)**
47. Vitamins **(eg. Nicotinic acid, thiamine, pyridoxine**
48. Amino acids **(eg. Arginine)**
49. Growth regulators **(eg. Auxins) .•**
50. Sterilization Methods Used in **Tissue Culture Laboratory**
51. . Transfer of explant (**root, stem, leaf, etc.**) on to a culture medium is Called **inoculation .**
52. **Flamed and cooled forceps** are used for transfer of plant materials toDifferent culture media kept in **glasswares**.
53. The culture medium with the inoculum is incubated **at 26 – 28oC wit**h The light intensity at **2000 to 4000 lux (unit of intensity of light)** and Allowing photoperiod of **16** hour of light and **8** hours of **darkness**.
54. The **callus** is an **unorganized** mass of undifferentiated tissue.
55. Formation of new organs from the callus under the influence of auxin And cytokinin is called **morphogenesis**.
56. Such embryos are called **somatic embryos** result in the formation ofYoung plantlet.**Two** Types of morphogenesi**s** • Organogenesis • Embryogenesis .
57. Formation of new organs such as shoot and root is **known asOrganogenesis**.
58. The development of shoot from the callus is called **caulogenesis** and Formation of root is called **rhizogenesis** respectively.
59. Embryos arise from somatic callus tissue and are called **somatic Embryos or embryoids or somaclonal embryos .**
60. Exposing the plantlets to the natural environment in a stepwise manner Is known as **hardening**
61. **6** type of tissue culture .
62. Presence of growth hormones in the culture media promotes **callus Formation and proliferation .**
63. In vitro callus provides **totipotent** cells for plant regeneration via Organogenesis or somatic embryogenesis
64. Incubate cultures at **22-24 ± 2°C in light or dark**
65. Most callus will be initiated from the cut surfaces **within 3-8 wks**
66. **.**Set up a growth Latin Square of **25** culture plates wit**h 5 each** of auxin And **cytokinin** conc.
67. Callus for this growth trial should be uniform and of large (**20 mg) Size.**
68. Subculture vigorous growing callu**s (2-5 Mm dia.)**
69. **Three** procedures used for cell dispersion:
70. Callus passaged on **a 7d cycle for 2-3 wks**
71. Ratio of **auxin/cytokinins altered, increase auxin** concentration
72. Us**e 2-3 g o**f friable callus per **100** ml of liquid medium
73. Subculture the cells to fresh medium at a **ratio of 1:1 .**

74. Initiation from callus treated with enzymes • **Pectinase • Cellulase**

75. Anthers or pollens can be cultured on a suitable medium containing sucrose (**usually 2%),** iron, vitamins, hormones etc.

76. Callus is formed from **pollen grains** on a medium supplemented wit**h yeast Extract** or casein **hydrolysate**.

77. Nicotiana tabacum optimal temperature for embryoid **growth is 25°C.**

78. Pre-treatment of anthers **at 3—10°C for 2—30 days** stimulates embryogenesis. **Wenzel ('77) o**bserved that buds of Secale cerale pretreated **at 6°C for 6—10 days** Develop embryoids.

79. In N. Tabacum if the buds are pre- treated **at 5°C for 72 ho**urs Than **58% an**thers produce embryoids.

80. Centrifugation of the anthers **at 3—5°C for** approximately **30 minutes** helps embryoid formation

81. According to Nitsch ('72) and Sunderland **('71)** anthers with uninucleate pollens Are suitable for culture.

82. Protoplasts • Protoplasts are **naked** plant cells without the cell wall, but they possess plasma Membrane and all other **cellular components.**

83. Tissue is **immersed in 1.0 M sucrose** until protoplasm shrunk away from their Enclosing cell wall (Plasmolysis).

84. Leaf segments with mixture A (**Macerozyme in manifold at pH 5.8)** are vacuumed Infiltrated **for 5 mins**, transferred to a water bath **at 25°C an**d subjected to slow Shaking

85. Washed **3 times with 13%** mannitol

86. Cells are then incubated with 'enzyme mixture B' (**Cellulase in mannitol solution At pH 5.4) f**or above **90 mins at 30°C**

87. Resulting in small clumps of cell, also known as micro colony, **within 1 to 3 weeks**

88. A seed plant **embryo** is part of a seed, consisting of precursor tissues for the

89. Leaves, stem and root as well as one or more cotyledons. The embryo of different developmental stages, formed within the female Gametophyte through sexual process, can be isolated aseptically from the.(**Embryo culture)**

90. Incubate the cultures **at 22-25°C under a 16 h** photoperiod of **2000** Lux luminous intensity.

91. **Two types of** somatic embryogenesis

92. Direct somatic embryogenesis • Indirect somatic embryogenesi . nduced Embryogenic determined cells' (**IEDC**).

93. **Germplasm are** living genetic resources such as seeds or tissues that are Maintained for the purpose of plant breeding, preservation, and other research Uses.

94. **Haploid** is the term used when a cell has half the usual number of chromosomes

95. Somaclonal variation is the variation seen in plants that have been produced by **Plant tissue culture**

96. **Gametoclonal variation h**as been defined as the variation among plants Regenerated from gametic cells in culture .

97. Nullisomic is a genetic condition involving the lack of both the Normal chromosomal pairs for a species **(2n-2)**

98. Transposable elements (Tes), also known as "**jumping genes"**

99. **Transformation** was first demonstrated in **1928 by Britis**h bacteriologist **Frederick Griffith**.

100. In **1944** this "transforming principle" was identified as being genetic by Oswald Avery, Colin MacLeod, and Maclyn McCarty

101. **Viruses** are natural vectors fo**r genetic engineering.**The microprojectile bombardment method

102. Was initially named as biolistics by its **inventor Sanford (1988).**

103. **Two types of plant** •partical bombardment technique was invented by John **Stanford in 1984 for introduction of D**NA

104. There are **two types** of microinjection systems; **constant flow System and pulsed flow system**

105. First synthesized in 1874, DDT's insecticidal action was discovered by the Swiss Chemist Paul **Hermann Müller in 1939**

106. •**bees** pollinate more tha**n 70%** of all crops •be **$9.6 billion:** offset by about **$40 billion i**n Increased agricultural production.

107. There are **two types** of pesticide poisoning: •Acute poisoning •Chronic poisoning .

108. In **1940**, he showed that "Growth substances applied appropriately would kill Certain broad-leaved weeds in cereals without harming the crops

Final Term Portion

109. **Final portion quiz**

110. •The North Central Weed Science Society (**NCWSS**)

111. •Rotate crops, particularly those with different life cycles (**e.g. winter annuals such as winter Wheat, perennials such as alfalfa, summer annuals such as corn or soybeans).**

112. •**plant stress**"**Stress** is usually defined as an external factor that exerts a disadvantageous influence on the Plant.

113. **two** Types of Stress•**Biotic stresses •A-biotic stresses•Abiotic stress i**s defined as the negative impact of non-living factors on the living organisms In a specific environment

114. •Types of A-biotic Stresses **morphological, physiological, Biochemical, and molecular changes**

115. •example of a biotic stress **Water/ drought stress, Heat stress, Cold stress, Light stress, Wind stress, Salinity stress, Heavy Metals**

116. •There are **two** types of water stresses **1.**Drought **2.**Flooding / Water logging.

117. •Effects of Heat Stress **Dehydration. Yellowing of plants and leaves; chlorophyll deteriorates.**

118. •Effects of Cold Stress **Cold stress decreases membrane fluidity.**

119. Wind Stress About **50**% of crop yield in Pakistan is effected by the wind stress

120. **Salt stress is** one of the major A-biotic stresse

121. •Some physiological damages also Occur due to salt stress that are• **Water deficiency •Ion cyto-toxicity •Osmotic stress ….**

122. The accumulation **of Na+ and Cl- ions** in the cells is very toxic in terms that these ions can **Influence the enzymatic action**

123. •□ **Heavy metals** are generally defined as metals with relatively high densities, atomic Weights, or atomic number

124. .•□ Essentials heavy metals:□ **Required for all kinds of plants. Cu – Photosynthesis**

125. •Beneficial heavy metals:**Required for specific plant groups. Ni – Activation of Enzymes**

126. •Non-Essentials heavy metals: **Cd, Hg, Cr etc** **•Biotic stress is stres**s that occurs as a result of damage done to an organism by other living Organisms, such as bacteria, viruses, fungi, parasites, beneficial and harmful insects, weeds, And cultivated or native plants .

127. plant affect by a disease called **host**

128. while organism that produces a disease is termed as **pathogens** .

129. citrus cancer **xanthomonas compestris pv citri .**

130. Viruses are **submicroscopic** entities capable of causing disease.

131. They are a piece of **nucleic Acid (genetic material) surrounded by a protein c**oat.

132. Plant viruses are made up of **two** components .

133. •**Bacteria** are amongst the microbes which benefit as well as harm the plants. **Pathogenic** Bacteria which belong to the genera **like Xanthomonas, Erwini**

134. •**Blight disease** is marked by the **chlorosis** of different **plant tissue**

135. **Fire blight** is one such disease found in **pear and apple Plants; Erwinia amylovora** is the bacteria that causes fire blight

136. **•Brown Spot** DiseaseThe causal organism of this disease is the bacterium **called Pseudomonas syringae**.

137. gray spots, cracking of leaves, brownish Yellow leaf borders, etc **Brown spot**

138. •**Crown Gall Disease** This disease is soil-borne and affects a variety of plants including roses, apples, cherries, Raspberries, etc. The disease is named so because of the **galls (abnormal outgrowths**) formed at crowns

139. .•**Fungi** are one of the living organisms that can cause plant disease and are the cause of about **Eighty-five percent** of all plant diseases. More **than 100,000** species of fungi have been Classified and include **molds, mildews, and mushrooms.**

140. •**BROWN PATCH** It is a turfgrass disease caused by different species of the **Rhizoctonia fungus.** Patches of **brown And yellow color appear** on the lawn in **irregular shapes**. It does not affect the roots and crown of The grass, so it is also known as **Foliar disease.**

141. •There are some fungal infections that affect the roots and barks of the plants. One such **Fungi is canker fungi**.

142. •**CLEMATIS WILT** It is caused by a fungus (**Phoma clematidina)**,

143. •**DOWNY MILDEW** This disease is caused by **Peronosporaceae** which affects a number of plants. It can be identified When discolored blotches appear on the leaves; a **mold-like growth**

144. •**CLUB ROOT** This infection is found in cabbages, turnips, and radishes which is caused by the **Plasmodiophora brassicae fungus**.

145. •**DAMPING OFF** This is a disease caused by different **fungi – Pythium and Phytophthora** being some of them

146. •Some insects can cause serious damage if **left untreated.** Boring insects, such as the clearwing **Moth, ambrosia beetle and other species of beetles**

147. •Beetle galleries can be very **extensive**

148. •Tricarboxylic Acid cycle [**TCA**]

149. •increased activity of phenol oxidases have been shown in disaeses caused by **phytophthora And Fusarium** species

150. •Formation of cork layer takes place after infection of certain disease [**scab of potato, soft root Of sweet potato]**

151. •**Tyloses** are outgrowths of the protoplast of adjacent parenchymatous cells protruded into

152. **Xylem vessels.** •Swelling of the outer wall of epidermal and sub epidermal cells takes place as **defense respon**

153. **Phaseolin** : Detected in bean leaves incubated with the bacterium **Pseudomonas phaseolicola**

154. **Necrotrophy**, in which the plant cells **are killed**

155. **Biotrophy**, in which the plant cells **remain alive**

156. •**Hemibiotrophy**, in which the pathogen initially keeps cells alive but kills them at later stages Of infection .

157. •pre-formed inhibitors are the **saponins** and the **glucosinolates**

158. •A biologically active **triterpinoid saponin** found in the **roots** of **oat plant**

159. •the production of ROS is often the first response detected, occurring **within 5 min** Superoxide and hydrogen peroxide (**H2O2**)

160. NO is known to bind **heme** and could inhibit catalase and ascorbate peroxidase, which **Detoxifies H2O2**

161. •Both SA and BA are derived from the **phenylpropanoid pathway** and have many roles in **plant Defense responses** •asmonic acid (JA) is an **oxylipin**-like hormone derived from oxygenated **linolenic acid**

162. •Ethylene is frequently synthesised during both **incompatible and compatible interactions**

163. •**Abiotic stresses** reduce average yield of crops by **upto50%**

164. •global loss due to insect pests in potential yields of all **crops is ~14%.**

165. •**Annually about 42%** of the crop productivity is lost owing to various abiotic stress factors.•• Properly acclimated plants can survive overwintering at extremely **low Ts** •

166. Temperature changes **lipid** and thus **membrane** propertie

167. •Biotic and abiotic stresses can reduce average plant productivity **by 65% to 87%,**

168. •**Oxidative stress re**sults from conditions promoting the formation of active oxygen species that **Damage or kill cells** • Hydrocarbons and oxides of nitrogen (NO, NO2) and sulfur (Sox) react with solar UV Radiation to **generate ozone (O3).**

169. **Ozone** is a highly reactive **oxidant**.

170. •Resistance to ozone Utilizes either **avoidance or tolerance**

171. •The ROS production in plants is mainly localized in the **chloroplast, mitochondria and peroxisomes.** Not **found in cell wall**

172. ROS) were initially recognized as **toxic by-products** of aerobic metabolism.

173. •The ROS family include **free radicals** lik**e O•−2, OH•**And Non-radicals like **H2O2 and 1O2**

174. **.•** There are **secondary sites** as well like the endoplasmic reticulum, cell membrane, cell wall and the Apoplast.

175. The ROS is being produced under **both normal and stressful conditions**

176. •in prese**nce of Light, chloroplasts and peroxisomes** are the major sources of ROS production, while the **Mitochondrion is the leading** producer of ROS under dark conditions .

177. •ROS is known to cause damages to biomolecules such **as lipids, proteins and DNA**

178. ROS in membrane phospholipids are the double bond between **C-atoms** and the ester linkage between **glycerol and fatty acids .**

179. •Lipids form a major portion of the **plasma membran**

180. •plant nuclear DNA is well protected by **histones** and a**ssociated proteins**,

181. both **Mitochondrial** and **chloroplastic** DNA bears the brunt of the ROS attack due to lack of Protective histones

182. •The ROS defense mechanism consists of the **antioxidant machinery** .

183. The antioxidant machinery has **two** arms with the **enzymatic** components and **non-enzymatic Antioxidants**

184. •SOD belongs to the family of **metalloenzymes** present in all **aerobic organisms**

185. •Peroxisomes are the **hotspots** of H2O2production due to β-oxidation of fatty acids,

186. •**Ascorbic acid (AA)** as a reducing agent

187. •**Carotenoids** belong to family of **lipophilic antioxidants** which are localized in the plastids of Both **photosynthetic and non-photosynthetic** plant tissues.

188. **Heat stress** is often defined as the **rise in temperature** beyond a threshold level for a period of time Sufficient to cause **irreversible** damage to plant **growth and development**

189. •**Two hormones** named as **Abscisic acid (ABA) and ethylene (C2H4), ar**e used as **stress hormones,**

190. **GMO** Genetically modified organism

191. •**GMO** is an organism whose genetic material has been altered using genetic engineering Techniques.

192. •first recombinant DNA molecules by **Paul Berg in 1972.**

193. •**MICROBES** Bacteria were the first organisms to be modified in the laboratory, due to their simple genetics.

194. •**MAMMALS** Ralph L. Brinster and Richard Palmiter developed the techniques responsible for transgenic mice, Rats, rabbits, sheep, and pigs in the early **1980s**

195. •Herbicides are chemicals used to **kill weeds.**

196. •**Cross Breeding**:This technique has been used **since the 1700s**

197. •humans, are **diploid** animals,

198. •hard cell walls Removed (**Protoplasts**)

199. •**Genome Editing** This process has the ability to cut, replace or insert genes within the seed cells using "molecular **Scissors" called nucleases—enzymes**

200. •**Transgenisis**: It occurs when genes from one crop are incorporated into another crop

201. •It is confirmed for the first Time in the world that a plant gene is functioning properly in a living mammal, not in a cultured **Cell," said by professor akira**

202. •they have short generation interval(**114 days) an**d their genome is quite similar to **Humans (3x times than mouse**).

203. •The ways by whic**h GMO toxicity** can affect animals, plants and soil are: **Cancer**, damage to native Species, they can pollute the environment, these can deplete soil minerals,

204. •Already **37** people have died from side effec**ts, 1,500** are partially paralyze**d, 5,000** were Temporarily handicapped by chemicals used in harvesting

205. •**Golden rice** is genetically modified rice that now contains a large amount **of A-vitamins .** •**Beta-carotene** gives carrots their **orange** .

206. **Three** new genes are implanted: two from **daffodils** and the third from a **bacterium**

207. •Long-lasting, genetically modified tomatoes came on to the **market in 1994**

208. •popularly Known as Bt genes) from bacterium **Bacillus thuringiensis**

209. •**Transgenic crops** (e.g., cotton, rice, maize, potato, tomato, brinjal, cauliflowers, Cabbage, etc.) •**Male sterile p**lants are very important to prevent **unnecessary pollination**

210. **.** •Since the **1980s several c**ompanies are aiming to develop genetically modified crops and to market Them. Chemical company Monsanto dominates the market for seeds of GM crops **Holding 70%–100% mark**et share for various crops.

211. Other companies**: Pioneer, Syngenta and Bayer**

212. •There ar**e 2 types** of Bt crops, such as **Bt corn and Bt cotton**, •

213. hTere is **9 percent** global primary crop production from genetically modified crops, Like in **2006: over 100 million hectares in 22 countries** were of GM crops in which 64% of total Were soya crop and **24**% are of the global maize crop

214. **85** percent of corn grown in the U.S

215. •Canola is genetically engineered form was **approved in 1996,** and as of **2006, around 90 percen**t Of U.S. canola crops are genetically modified

216. •It has been estimated that **40-250 kg N / ha /** year is fixed by different legume crops by The microbial **activities of Rhizobium.**

217. **Azospirillum**- It belongs to bacteria and is known to fix the considerable quantity of nitrogen In the **range of 20- 40 kg N/ha in** the rhizosphere in non- non-leguminous plants such as cereals, Millets, Oilseeds, cotton etc.

218. •**Cyanobacteria**- A group of one-celled to many-celled **aquatic organisms.** Also known as **blue Green algae**

219. Amount of CTAB bufer used for extraction of DnA-------- 1%,

220. Most abundant hormone in plants---------- **Auxin**

221. Incubation of callus culture ---------- 26 - 28oC with the light intensity at 2000 to 4000 lux (unit of intensity of light) and allowing photoperiod of 16 hour of light and 8 hours of darkness.

222. Plastid genome is---------- Haploid

223. Anthesis is---------- **flowering period of plants**

224. Who created the first recombinent DNA---------- **Paul Berg in 1972.**

225. which possess half the number of chromosomes from a Diploid species.___**monoploids**

226. It has been estimated that __ is fixed by different legume crops by the microbial activities Of Rhizobium.__**40-250 kg N / ha / year**

227. Worldwide, there is percent global primary crop production from genetically modified crops,___**9%**

228. Canola is genetically engineered form was approved in___ **1996**,

229. injecting a gene for a rat Growth hormone into a mouse embryo in___ **1982**.

230. GMO stand for ___ **genetically modified organisms**

231. Transformed cells with a single copy of the transgene usually show higher and

232. More stable expression than multicopy lines, in which expression may suffer from Posttranscriptional ___ **gene silencing.**

233. Rate of transportation _____ in plant with pathogenic attack ___ **increase**

234. Hormone induce callus formation ___ **cytokinin**

235. The genetic markers should be_____and_____: **palymorphic**;. **Heterazygous**:

236. Almost perecent of corn grown in the U.S. is genetically modified.__**85%**

237. The creation of the first recombinant DNA molecules by Paul Berg in_____ **1972**

238. _____ can usefully be defined as the branch of philosophy concerned witecide what is Morally wrong and what is morally right. ___**Ethics**

239. Glycine is___ acid ___ **amino acid**

240. Which possess the half no of chromosomes from diploid species (**monoploid**)

241. For tissue growth in the medium sucrose concentration of **(2%)**

242. Almost (**85**) of corn grown in U.S is genetically modified

243. Rate of respiration (**increases**) in diseased plants

244. Blight disease is caused by (**chlorosis**)

245. It has been estimated that (**40-50 kg N/ ha/ year) i**s fixed by different legumes crops by microbial activity

246. In (India) technique of cyanobacteria inoculation to rice field being followed

247. Russian study of rat fed with GMO soy show (**five**) folds increase in mortality
248. (**GM**) Tomato can tolerate lengthier shelf life
249. RNA hydrolyzing enzymes (Such sterile male plants are created by introducing a gene coding for an Enzyme (**barnase**), which is an RNA hydrolyzing enzyme) that inhibits Pollen formation.)
250. Chilling injuries common in plant Except (**Maize**)
251. (**Ammonium nitrate) Is** macronutrient for Murashige and skoog
252. Auxin Favor DNA duplication and Cytokinin enables the separation of (**chromosomes separation**)
253. Growth Regulators/ growth promotors hormones (**Auxins, Gibberellins, Cytokinin)**
254. How pesticides enter our body by (**mouth, lungs, skin) (2 times**)
255. In 1990 1st food product of biotechnology is (**Cheese**)
256. Most callus will be initiated cut surface within(**3 to 8 weeks**)
257. Heat shock protein size of HSP100 class is (**100 to 114 Kda)**
258. Which cause ripening (**ethylene**)
259. How many phases of regeneration? (**Three**)
260. Multiple form of same enzymes coded by different genes (**Isozymes**)
261. Powdery mildew is (**fungal disease)**
262. Commonly transformation occur in (**bacteria**)
263. ROS is not found in….(The ROS production in plants is mainly localized in the chloroplast, Mitochondria and peroxisomes.)
264. Late blight of potato disease caused by (**phytopthora infestants)**
265. Plastid genome _____ **Haploid**
266. Light stress is __**a-biotic**
267. Incorporation of genes from one to another _**Transformation**__
268. Herbiside used where _**weeds**__
269. .Types of genration **_two**__
270. Which is firstly used for genetic **technique Bacteria____**
271. Which is work at moderate temperature and help in cells formation ___
272. Vessels are**_glasswares**___
273. .Ozone cause **all of these**
274. Optimum temperature for culture media **_22 to 28 degree C__**
275. Apoenzyme are……..**the protein part of an enzyme**
276. symptom of pesticides which are not for short time……
277. vessels used for culture **petri plates, test tubes, "Magenta boxes," bottles, and flasks…** (lead to prevention water & mineral Translocation)
278. herbicide are…….**chemicals used to kill weeds.**
279. Tumer causing bacteria……**Helicobacter pylori**
280. . FR13A rice-type has high level flooding **Tolerance**
281. Wind Stress About **50%** of crop yield in Pakistan
282. Somaclonal variation is the variation seen in plants that have been produced by **plant tissue culture**

283. In **1907** a bacterium that caused plant tumors, **Agrobacterium tumefaciens**,

284. blight potato disease caused by------ **fungus Phytophthora**

285. How do pesticides enter our bodies?- **through the skin (dermal) through the lungs (inhalation) by mouth (ingestion).**

286. **Isozyme markers:** Multiple forms of the same enzyme coded by the different genes.

287. Pigs and cows became **sterile** from GM corn.

288. There are **two** types of microinjection systems; **constant flow system and pulsed flow system**

289. Microinjection • where the DNA is directly injected into plant protoplasts or cells (specifically into the nucleus or cytoplasm) using fine tipped **(0.5 - 1.0 micrometerdiameter)** glass needle or micropipette.